Roll Number: _____

Thapar Institute of Engineering & Technology, Patiala

Department of Computer Science and Engineering

End Semester Examination (Summer)

B.E. -Summer Sem	Course Code: UCS310 Course Name: Database Management Systems
Date: July 18 , 2026	Time: 9:00
Duration: 3 Hours, M. Marks: 40	Name of Faculty: Dr. Harpreet Singh

Note: Attempt any 5 questions in order. Assume missing data, if any, suitably.

		Marks	CO/ BL
Q1	Consider the following four transactions: T1 = {$W_1(A)$, $W_1(B)$}, T2 = {$W_2(A)$, $R_2(C)$, $W_2(C)$}, T3 = {$R_3(A)$, $W_3(B)$, $R_3(C)$, $W_3(C)$}, and T4 = {$R_4(B)$, $R_4(C)$, $W_4(A)$}. The following schedule is executed: S = $W_1(A)$, $W_2(A)$, $R_3(A)$, $W_1(B)$, $R_4(B)$, $W_3(B)$, $R_2(C)$, $W_3(C)$, $W_2(C)$, $R_3(C)$, $R_4(C)$, $W_4(A)$. Draw the precedence graph for the given schedule and determine whether it is conflict serializable, providing appropriate justification. Next, determine whether the schedule is view serializable. If the schedule is view serializable, specify the equivalent serial schedule(s); otherwise, justify why no equivalent serial schedule exists.	(8)	4/2
Q2	Consider two transactions T1 and T2 operating on data items A, B, and C. Transaction T1 performs the operations R(A), W(A), R(B), W(B), Commit, while transaction T2 performs R(B), W(B), R(C), W(C), Commit. Assume that the initial values of the data items are A = 100, B = 200, and C = 300, where T1 increments A by 20 and decrements B by 30, and T2 increments B by 50 and doubles the value of C. The proposed execution schedule is S = $X_1(A)$, $R_1(A)$, $W_1(A)$, $X_2(B)$, $R_2(B)$, $W_2(B)$, $X_1(B)$, $R_1(B)$, $W_1(B)$, $X_2(C)$, $R_2(C)$, $W_2(C)$, C_1, C_2, where X denotes an exclusive lock and, the symbols C_1 and C_2 denote the commit operations. Assuming that the Rigorous Two-Phase Locking protocol is followed, simulate the execution of the schedule. Draw the lock table after each lock request, determine whether a deadlock occurs, and justify your answer. Finally, compute the final values of A, B, and C after the successful execution of both transactions.	(8)	4/2
Q3	A database system uses the Undo/Redo Recovery technique to recover from system failures. The initial balances of three bank accounts are A = ₹1000, B = ₹2000, and C = ₹3000. Two transactions are executed concurrently. Transaction T1 transfers ₹300 from account A to account B, while Transaction T2 deposits ₹500 into account C and then withdraws ₹200 from account B. The system generates the following log records: <START T1>, <T1, A, 1000, 700>, <T1, B, 2000, 2300>, <COMMIT T1>, <START T2>, <T2, C, 3000, 3500>, <CHECKPOINT>, <T2, B, 2300, 2100>, after which the system crashes before T2 commits. Assuming that the database uses the Undo/Redo Recovery algorithm, identify the transactions that should be included in the Redo List and Undo List, perform the necessary recovery operations in the correct order, and determine the final values of accounts A, B, and C after recovery.	(8)	4/3
Q4	A university is developing a Course Management Information System to monitor student enrollments, faculty assignments, classroom utilization, and course delivery so that academic activities can be managed efficiently and institutional resources can be used effectively. To support this system, the university maintains course-related information in the following relation: COURSE_MANAGEMENT (StudentID, StudentName, FacultyID, FacultyName, Department, RoomNo, RoomType, CourseID, CourseName, Credits, Semester, TeachingAssistantID, TeachingAssistantName) The following functional dependencies are known: StudentID → StudentName FacultyID → FacultyName, Department RoomNo → RoomType CourseID → CourseName, Credits	(8)	3/2

	TeachingAssistantID → TeachingAssistantName (StudentID, CourseID, Semester) → FacultyID, RoomNo, TeachingAssistantID (a) Determine super key and candidate keys of the relation with proper justification. (b) Normalize the relation up to Fourth Normal Form, clearly showing each decomposition step. Determine whether the resulting 3NF relations satisfy BCNF.		
Q5	(a) Explain the use of ANY and ALL operators. (b) Describe the use and limitations of check constraint. (c) Explain shadow paging. Give advantages and disadvantages of shadow paging technique	2+3+3	5/2
Q6	Consider the following two relations: Employee (Emp_ID, Emp_Name, Dept_ID, Salary) and Department (Dept_ID, Dept_Name). The Employee relation contains the tuples (101, Amit, 10, 50000), (102, Priya, 20, 60000), (103, Rohan, 30, 55000), (104, Neha, 40, 65000), (105, Karan, NULL, 48000), while the Department relation contains the tuples (10, HR), (20, Finance), (30, IT), (50, Marketing). Based on these relations, write the output of the following SQL queries: (i) SELECT E.Emp_Name, D.Dept_Name FROM Employee E INNER JOIN Department D ON E.Dept_ID = D.Dept_ID; (ii) SELECT E.Emp_Name, D.Dept_Name FROM Employee E LEFT OUTER JOIN Department D ON E.Dept_ID = D.Dept_ID; (iii) SELECT E.Emp_Name, D.Dept_Name FROM Employee E RIGHT OUTER JOIN Department D ON E.Dept_ID = D.Dept_ID; and (iv) SELECT E.Emp_Name, D.Dept_Name FROM Employee E FULL OUTER JOIN Department D ON E.Dept_ID = D.Dept_ID;. Assume that the database management system supports FULL OUTER JOIN. Write the result of each query in tabular form.	8	5/3